

ANSWER KEY					AT# 09 (ELECTROSTATICS)				
Q.	1	2	3	4	5	6	7	8	9
A.	1	1	4	2	2	3	1	4	1
Q.	10	11	12	13	14	15	16	17	18
A.	1	2	1	1	4	4	4	3	1
Q.	19	20	21	22	23	24	25	26	27
A.	3	1	3	2	2	3	1	4	3
Q.	28	29	30	31	32	33	34	35	36
A.	2	3	3	2	3	3	1	1	1
Q.	37	38	39	40	41	42	43	44	45
A.	1	2	3	3	4	3	1	4	4

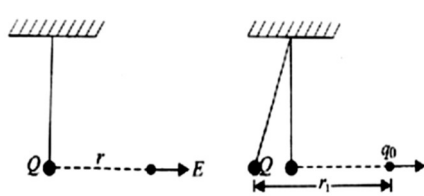
Hints And Solution

1. ANS - 1

Figure indicates the presence of some positive charge to the left of A

$$\therefore E_A > E_B \quad (\because r_A < r_B)$$

2. ANS - 1



$$E = \frac{KQ}{r^2} \text{ --- 1}$$

$$F = \frac{kQq_0}{r_1^2} = \frac{F}{q_0} = \frac{kQ}{r_1^2} \text{ --- 2}$$

As $r_1 > r$, so from equation (i) and (ii), we get

$$E > \frac{F}{q_0}$$

3. ANS - 4

$$E = E_C - E_D - E_A \cos \theta - E_B \cos \theta$$

$$\frac{K(2Q)}{b^2} - \frac{KQ}{b^2} = 2 \times \frac{Kqb}{(b^2 + 4b^2)^{3/2}}$$

$$Q = \frac{2q}{5\sqrt{5}} \Rightarrow \frac{q}{Q} = \frac{5\sqrt{5}}{2}$$

4. ANS - 2

$$E = -\frac{dV}{dx} = -\frac{d(4x^2)}{dx}, E = -8x$$

5. ANS - 2

Ratio of potential of the bigger drop to smaller drop is given by,

$$\frac{V}{V_0} = n^{\frac{2}{3}} = 1000^{\frac{2}{3}} = 100$$

6. ANS - 3



$$V_1 = \frac{\sigma_1 r}{\epsilon_0} = \frac{\sigma_1 \times 2}{\epsilon_0}$$

$$V_2 = \frac{\sigma_2 R}{\epsilon_0} = \frac{\sigma_2 \times 3}{\epsilon_0}$$

As the spheres are charged to same potential,

$$\therefore \frac{\sigma_1 \times 2}{\epsilon_0} = \frac{\sigma_2 \times 3}{\epsilon_0}$$

$$\therefore \frac{\sigma_1}{\sigma_2} = \frac{3}{2}$$

7. ANS - 1

Proton and electron move under the influence of their electrostatic attraction. Therefore Momentum is conserved i.e constant; hence

$K = \frac{p^2}{2m}$. As electron has less mass it possesses great kinetic energy.

8. ANS - 4

We know from the application of Gauss's law that a conductor is always equipotential.

Therefore, potential at centre and potential at one corner is the same. The ratio of potential is 1: 1

9. ANS - 1

Position of null point is given by

$$x_1 = \frac{r}{\sqrt{\frac{q_2}{q_1} + 1}}$$

$$x_1 = \frac{30}{\sqrt{\frac{4}{1} + 1}}$$

$$x_1 = 10\text{cm}$$

10. ANS - 1

$$V_A - V_B = - \int_B^A \vec{E} \cdot d\vec{r}$$

$$V_A - V_B = - \int_{2r}^r \frac{2\lambda}{4\pi\epsilon_0 r} dr$$

$$V_A - V_B = \frac{2\lambda}{4\pi\epsilon_0} \int_r^{2r} \frac{1}{r} dr$$

$$= \frac{2\lambda}{4\pi\epsilon_0} \ln \frac{2r}{r}$$

$$= \frac{2\lambda}{4\pi\epsilon_0} \ln 2$$

11. ANS - 2

The electrical potential of a dipole is given by,

$$V = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \vec{r}}{r^2}$$

At equatorial plane $\theta = \pi/2$, therefore, electric potential at any point in the equatorial plane of a dipole is zero.

12. ANS - 1

Frequency of time period of SHM is dependent on variable forces. It does not depend on constant external force. Constant external force can only change the mean position. For example, in the given question mean position is at natural length of spring in the absence of electric field. Where is in the presence of electric field, mean position will be obtained after a compression of x_0 . Where x_0 is given by

$$K x_0 = QE$$

$$\text{Or } x_0 = QE/K$$

Correct answer is (of the same frequency and with shifted mean position)

13. ANS - 1

Potential difference between two point in electric field is

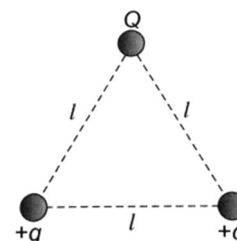
$$V_A - V_B = W/q_0$$

Where W is the work done by moving charge q_0 from point A to B.

So,

$$V_A - V_B = \frac{2}{20} = 0.1V \quad (\text{Here } W=2J, q_0=20C)$$

14. ANS - 4



Electrostatic potential energy of the system of charges is

$$U = \frac{1}{4\pi\epsilon_0} \left[\frac{Qq}{l} + \frac{Qq}{l} + \frac{q^2}{l} \right]$$

$$U = \frac{1}{4\pi\epsilon_0} [2Qq + q^2]$$

Given, $U = 0$

$$2Qq + q^2 = 0$$

$$Q = -\frac{q}{2}$$

15. ANS - 4

$$E_{\text{east-west}} = 0 \quad \left(\frac{dV}{dr} \right)_{\text{east-west}} = 0$$

16. ANS - 4

$$\sigma = 26.5 \mu\text{Cm}^{-2} = 26.5 \times 10^{-6} \text{Cm}^{-2}$$

$$\frac{dF}{dS} = \frac{\sigma^2}{2k\epsilon_0}$$

$$= \frac{26.5 \times 26.5 \times 10^{-12}}{2 \times 1 \times 8.85 \times 10^{-12}} = 39.67 \text{N/m}^2$$

17. ANS - 3

The electric field of an electric dipole at a point on its axis at a distance d from the centre of the dipole is

$$E = \frac{2p}{4\pi\epsilon_0 d^3}$$

where p is the dipole moment of the dipole.

$$\text{Hence } E \propto \frac{1}{d^3}$$

18. ANS - 1

Since surface charge densities are equal, Q is divided in the ratio of surface areas

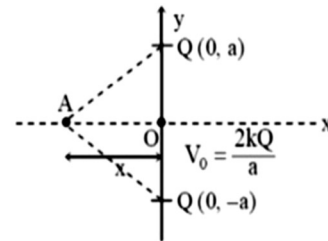
$$Q_r = \frac{r^2 Q}{r^2 + R^2} \quad Q_R = \frac{R^2 Q}{r^2 + R^2}$$

$$\therefore V = \frac{1}{4\pi\epsilon_0} \left[\frac{Q_r}{r} + \frac{Q_R}{R} \right] = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{r^2 + R^2} [r + R]$$

19. ANS - 3

Lithium nucleus and carbon nucleus are positively charge. According to coulomb law, same charge repel each other. So, nuclei do not come very close.

20. ANS - 1



The potential at the curve is $V_0 = \frac{2kQ}{a}$

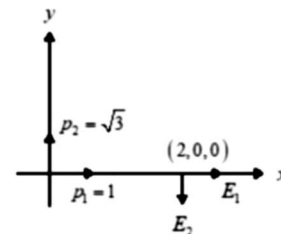
Potential on the x-axis is given by,

$$V_A = \frac{2kQ}{\sqrt{a^2 + x^2}}$$

$$\text{At } x = \infty, V_\infty = 0$$

So, the graph will tend to 0 at $x = \pm\infty$ and will be maximum at $x = 0$

21. ANS - 3



$$\text{Let } \vec{p} = \hat{i} + \sqrt{3}\hat{j} = \vec{p}_1 + \vec{p}_2$$

$$\text{Therefore } \vec{p}_1 = \hat{i} \text{ and } \vec{p}_2 = \sqrt{3}\hat{j}$$

Field due to $\vec{p}_1$ at point B

$$E_1 = \frac{2 \times K}{2^3} = \frac{K}{4}$$

Field due to $\vec{p}_2$ at point B

$$E_2 = \frac{\sqrt{3} K}{8}$$

$$E_{\text{net}} = \sqrt{E_1^2 + E_2^2}$$

$$E_{\text{net}} = \sqrt{\frac{k^2}{16} + \frac{K^2 \times 3}{64}}$$

$$E_{\text{net}} = \frac{\sqrt{7} K}{8}$$

22. ANS - 2

Electric Potential

$$V = \left(\frac{1000}{x} + \frac{1500}{x^2} + \frac{500}{x^3} \right)$$

Electric field

$$E = -\frac{dV}{dx} = \left(\frac{1000}{x^2} + \frac{(2)1500}{x^3} + \frac{(3)500}{x^4} \right) \hat{i}$$

At $x=1$

$$E = 5500 \hat{i} \text{ V m}^{-1}$$

23. ANS- 2

When force due to the gravitational field is equal to force due to the electric field, the particle must follow a straight-line path with acceleration.

When force due to the gravitational field is not equal to force due to the electric field, the particle will follow the path of parabola. Hence 1 & 3 are correct

24. ANS - 3

(1) Suppose a point where $E=0$ between two positive charges but net potential is the scalar sum of potential of two charges.

(2) Suppose a point where $V=0$ between one positive and one negative charge but net electric field is in the direction from positive to negative charge

(3) Suppose a point between one positive and one negative charge, where $E \neq 0$ but net potential may be zero depending on the location of point between them

(4) Suppose a point where $V \neq 0$ between two positive charges but we cannot say that E cannot be zero because at neutral point we must say that $E=0$

25. ANS - 1

Excess of pressure inside liquid bubble

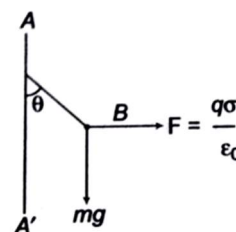
$$P_1 = \frac{4T}{R}$$

Electrostatic pressure due to charge $P_2 = \frac{\sigma^2}{2\epsilon_0}$

Therefore,

$$T = \frac{\sigma^2 R}{8\epsilon_0}$$

26. ANS - 4



The diagram is as follows

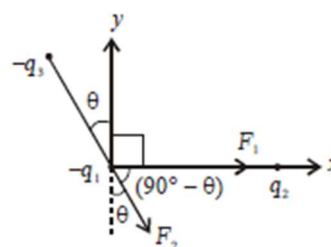
The electric field due to charged infinite conducting sheet is $E = \frac{\sigma}{\epsilon_0}$

Now force on the charged ball is $F = qE = \frac{\sigma}{\epsilon_0} q$

The resultant of electric force and mg balance the tension produced in the string

$$\tan \theta = \frac{F}{mg} = \frac{\frac{\sigma}{\epsilon_0} q}{mg} = \frac{q\sigma}{\epsilon_0 mg}$$

27. ANS - 3



$$\text{Force on } -q_1 \text{ due to } q_2 = \frac{q_1 q_2}{4\pi\epsilon_0 b^2}$$

$$\therefore F_1 = \frac{q_1 q_2}{4\pi\epsilon_0 b^2} \text{ along } (q_1 q_2) \text{ s}$$

$$\text{Force on } -q_1 \text{ due to } -q_3 = \frac{q_1 q_3}{4\pi\epsilon_0 a^2}$$

$$\therefore F_2 = \frac{q_1 q_3}{4\pi\epsilon_0 a^2} \text{ as shown}$$

F_2 makes an angle $(90^\circ - \theta)$ with F_1

Resolve part of F_2 along F_1

$$= F_2 \cos(90^\circ - \theta) = F_2 \sin \theta = \frac{q_1 q_3}{4\pi\epsilon_0 a^2} \sin \theta$$

Total force on $-q_1$

$$= \left(\frac{q_1 q_2}{4\pi\epsilon_0 b^2} + \frac{q_1 q_3}{4\pi\epsilon_0 a^2} \sin \theta \right) \text{ along } x \text{ axis}$$

The x component of force is proportional to

$$\left(\frac{q_2}{b^2} + \frac{q_3}{a^2} \sin \theta \right)$$

28. ANS - 2

$$\frac{1}{2}mv^2 = q\Delta v$$

$$\frac{1}{2}mv^2 = q \times \frac{\lambda}{2\pi\epsilon_0} \left(\ln \left(\frac{r}{r_0} \right) \right)$$

$$v^2 = \frac{q\lambda}{2\pi\epsilon_0} \ln \left(\frac{r}{r_0} \right)$$

$$v = \sqrt{\frac{q\lambda}{2\pi\epsilon_0} \ln \left(\frac{r}{r_0} \right)}$$

29. ANS - 3

$$\tau = PE \sin 30$$

$$10\sqrt{3} = \frac{PE}{2}$$

$$PE = 20\sqrt{3}$$

$$\text{Potential energy} = -PE \cos 30^\circ$$

$$= -20\sqrt{3} \times \frac{\sqrt{3}}{2}$$

$$= -10 \times 3 = -30 \text{ J}$$

30. ANS - 3

When drop is stationary, then $q_1 E =$

$$6\pi\eta r v_0 \text{ or } q_1 = 6\pi\eta r v_0 \frac{1}{E}$$

When drop moves upwards, then

$$3q = 6\pi\eta r (v_0 + v) \frac{1}{E} = \frac{2 \times 6\pi\eta r v_0}{E} = 2q_1$$

$$q_1 = \frac{3}{2}q$$

31. ANS - 2

For potential of two spheres to be same,

$$\frac{kq_1}{r_1} = \frac{kq_2}{r_2}$$

$$\text{or } \frac{q_1}{r_1} = \frac{q_2}{r_2}$$

$$\frac{q_1}{6400 \times 10^3} = \frac{1 \times 10^{-6}}{10^{-2}}$$

$$q_1 = 640C$$

32. ANS - 3

Electric field is zero everywhere inside a metal (conductor) i.e., field lines do not enter a metal plus these are perpendicular to a metal surface.

33. ANS - 3

By Coulomb's law, when charge Q is divided into two charges q and $Q-q$

$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{q(Q-q)}{r^2} = \frac{1}{4\pi\epsilon_0} \cdot \frac{(Qq - q^2)}{r^2}$$

$$\text{For } F_{\max}, \quad \frac{dF}{dq} = 0 = (Q - 2q) = 0$$

$$\Rightarrow q = \frac{Q}{2} \text{ for the maximum force}$$

34. ANS - 1

Work is equal to the potential energy of the system

$$U = 4 \frac{k(q)(-q)}{a} + 2 \frac{k(q)(q)}{a\sqrt{2}}$$

$$U = \frac{kq^2}{a}(-4 + \sqrt{2})$$

$$U = \frac{q^2}{\pi\epsilon_0 a}(-1 + \frac{\sqrt{2}}{4}) = -\frac{0.65q^2}{\pi\epsilon_0 a}$$

35. ANS - 1

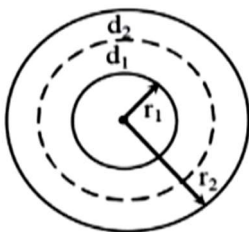
$$E_{axis} = \frac{1}{4\pi\epsilon_0} \frac{qz}{(R^2 + z^2)^{3/2}}$$

$$\Rightarrow \text{at } z = \frac{R}{\sqrt{2}}, \quad E_{axis} = \frac{2}{3\sqrt{3}} \frac{1}{4\pi\epsilon_0} \frac{q}{R^2}$$

$$\Rightarrow E_{axis} = \frac{2}{3\sqrt{3}} \cdot \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda \cdot 2\pi R}{R^2}$$

$$\Rightarrow E_{axis} = \frac{\lambda}{3\sqrt{3}\epsilon_0 R}$$

36. ANS - 3



Electric field,

$$r_1 < r, E = 0$$

$$r_1 < r < r_1 + d, \quad E = \frac{Q}{4\pi\epsilon_1 r^2}$$

$$r_1 + d < r < r_2, \quad E = \frac{Q}{4\pi\epsilon_2 r^2}$$

$$r > r_2, \quad E = 0$$

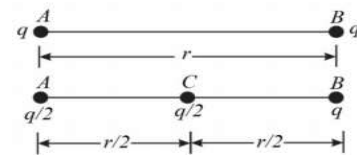
Potential can be finding out by integrating

$$\text{i.e., } V = - \int_{\infty}^r E \cdot dr = 0$$

37. ANS - 1

The induced charges will appear at the inner and outer surface of the metallic spherical shell. Also there is no charge in between the metal of the shell. So, the normal component of the electric field at any point on the Gaussian surface with radius between R_1 and R_2 will be zero.

38. ANS - 2



$$\text{Initially, } F_{AB} = \frac{1}{4\pi\epsilon_0} \cdot \frac{qq}{r^2} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q^2}{r^2}$$

$$\text{Finally, Force on C } F_C = F_{CB} - F_{CA}$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{\left(\frac{q}{2}\right)(q)}{\left(\frac{r}{2}\right)^2} - \frac{1}{4\pi\epsilon_0} \cdot \frac{\left(\frac{q}{2}\right)\left(\frac{q}{2}\right)}{\left(\frac{r}{2}\right)^2} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q^2}{r^2}$$

$$\text{So, } F_C = F_{AB} = F$$

39. ANS - 3

Speed obtain by the particle after falling through a potential difference of V volt is

$$v_A = \sqrt{\frac{2Vq}{m}} \text{ --- 1}$$

$$\text{And, } v_B = \sqrt{2V \times \frac{4q}{m}} \text{ --- 2}$$

Now dividing eq (i) by eq (ii), we get

$$\frac{v_A}{v_B} = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

So $v_A : v_B = 1 : 2$

40. ANS – 3

$$\text{Field due to small sphere} = \frac{K\rho_1 4\pi R^3}{(2R)^2}$$

Field due to bigger sphere will be

$$= \frac{K\rho_2 4\pi(2R)^3 \times R}{(2R)^3}$$

Equating above two, we get $\frac{\rho_1}{\rho_2} = 4$

41. ANS – 4

$$W_{CRD} = Q(V_D - V_C)$$

$$W_{CRD} = Q[V_D - 0]$$

($\because$ potential at centre of dipole is zero)

$$W_{CRD} = \left[\frac{kq}{3L} - \frac{kq}{L} \right] Q$$

$$W_{CRD} = -\frac{kqQ}{L} \times \frac{2}{3} = -\frac{qQ}{6\pi\epsilon_0 L}$$

42. ANS – 3

$$\tan \theta = \frac{V_y}{V_x}$$

$$\tan \theta \propto \frac{q}{m}$$

$$\frac{\tan \theta_1}{\tan \theta_2} = 2$$

43. ANS – 1

$$V_A - V_B = \int_a^b E \cdot dr = E(b - a) = \frac{\sigma r}{2\epsilon_0}$$

44. ANS – 4

From the conservation of mechanical energy

$$\frac{1}{2}mv^2 = \frac{kqQ}{r}$$

In second case

$$\frac{1}{2}m(2v)^2 = \frac{kqQ}{r'}$$

$$r' = \frac{r}{4}$$

45. ANS – 4

Charge on an isolated cube remains conserved so it remains neutral.